

Compton effect



Arthur Holly Compton

- 1892 – 1962
- American physicist
- Director of the lab at the University of Chicago
- Discovered the Compton Effect
- Shared the Nobel Prize in 1927



The Compton Effect, Introduction

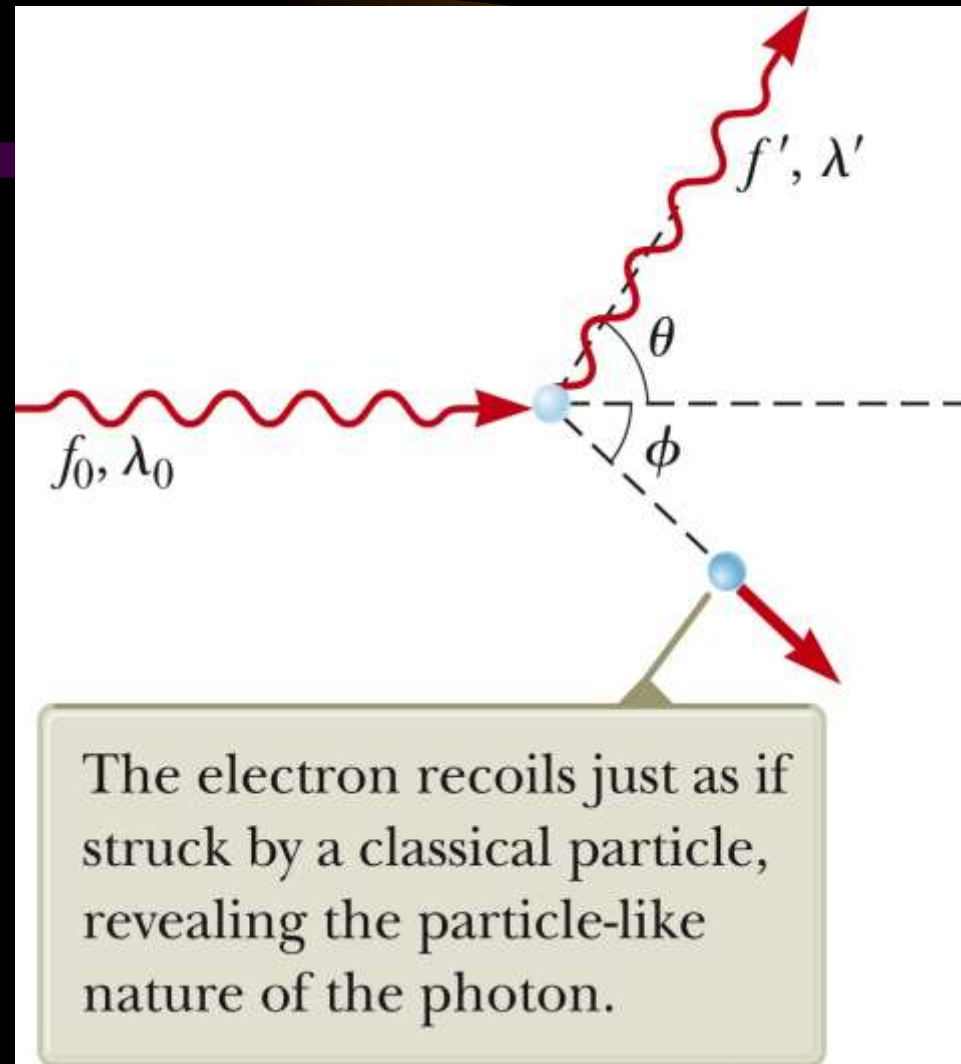
- Compton and Debye extended Einstein's idea of photon momentum.
- The two groups of experimenters accumulated evidence of the inadequacy of the classical wave theory.
- The classical wave theory of light failed to explain the scattering of x-rays from electrons.

Compton Effect, Classical Predictions

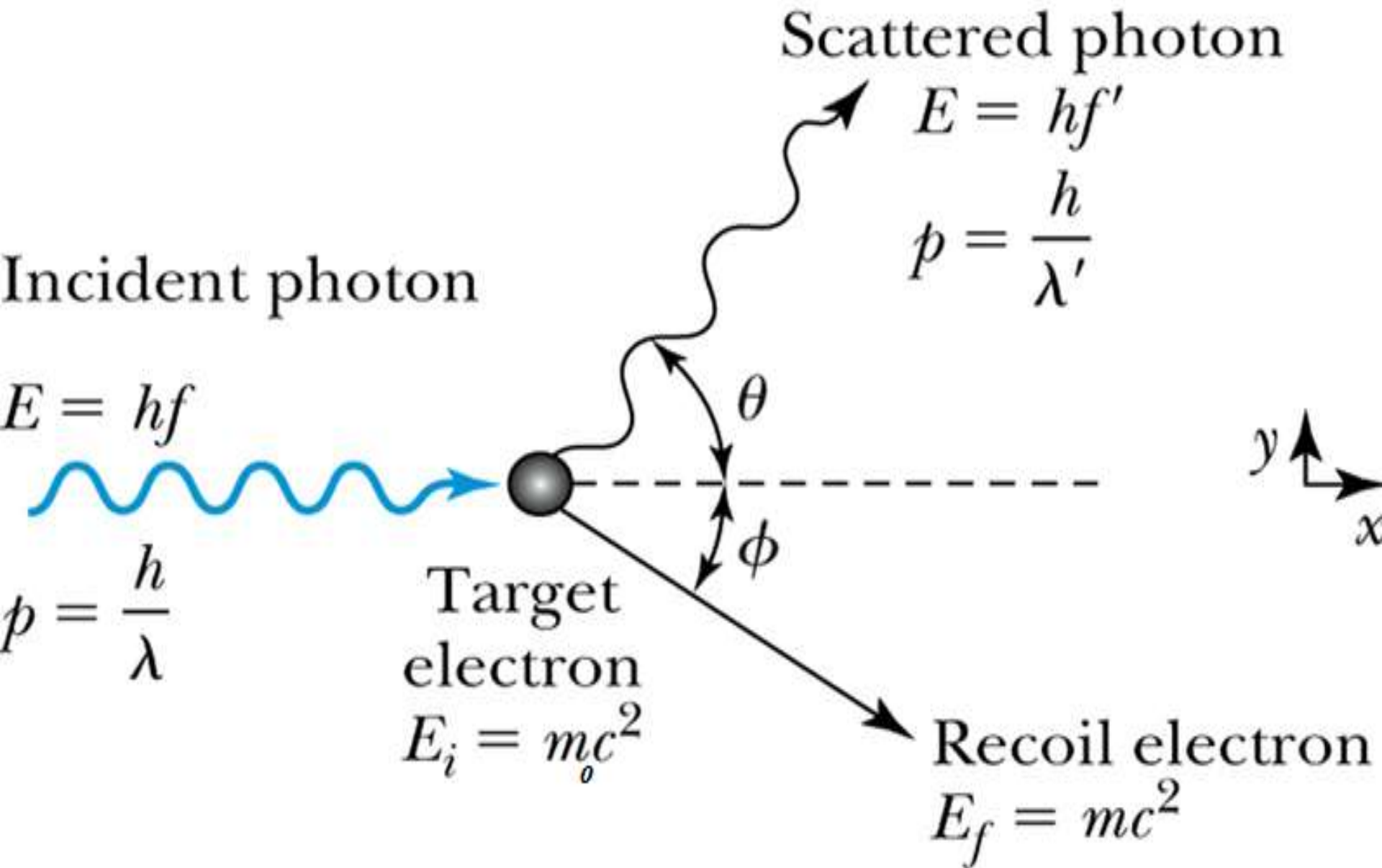
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- According to the classical theory, em waves incident on electrons should:
 - Have radiation pressure that should cause the electrons to accelerate
 - Set the electrons oscillating
 - There should be a range of frequencies for the scattered electrons.

Compton Effect, Observations

- Compton's experiments showed that, at any given angle, only *one* frequency of radiation is observed.



X-ray scattering: The Compton experiment

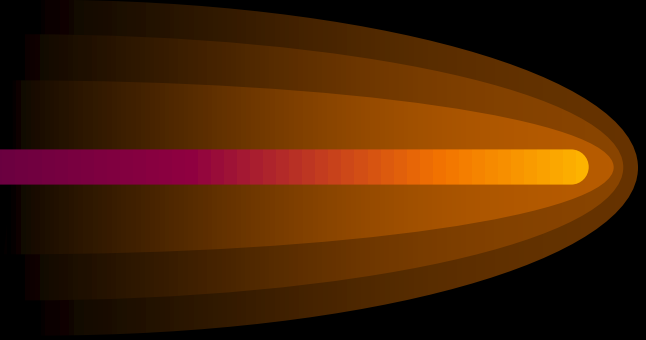


X-ray scattering: The Compton experiment

- In the Compton experiment, x rays are scattered from electrons. The scattered x rays have a longer wavelength than the incident x rays, and the scattered wavelength depends on the scattering angle ϕ .
- Explanation: When an incident photon collides with an electron, it transfers some of its energy to the electron. The scattered photon has less energy and a longer wavelength than the incident photon. These are elastic collisions, so both momentum and energy are conserved.

Compton Effect, Classical Predictions

- **Before collision**
- Energy of photon $= hf$
- Momentum of photon $p = \frac{hf}{c}$
- Energy of electron $= m_o c^2$
- Momentum of electron $p = 0$
- **After collision**
- Energy of photon $= hf'$
- Momentum of photon $p = \frac{hf'}{c}$
- Angle of photon $= \theta$
- Energy of electron $= m c^2$
- Momentum of electron $p = p$
- Angle of electron $= \Phi$



- According to law of conservation of energy

- $hf + m_0 c^2 = hf' + m c^2$

- $hf - hf' = m c^2 - m_0 c^2 \longrightarrow 1$

- According to law of conservation of momentum

- $\frac{hf}{c} + 0 = \frac{hf}{c} \cos\theta + p \cos\Phi \longrightarrow 2 \text{ Horizontal}$

- $0 + 0 = \frac{hf}{c} \sin\theta - p \sin\Phi \longrightarrow 3 \text{ Vertical}$

- Using equation 1, 2 & 3 we get

- $\frac{1}{f'} - \frac{1}{f} = \frac{h}{m_0 c^2} (1 - \cos\theta) \longrightarrow 4$

- $\lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos\theta) \quad \therefore \lambda = \frac{c}{f}$

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{mc} (1 - \cos\theta)$$

Compton Wavelength

- The factor $h/m_0 c$ in the equation is called the **Compton wavelength** of the electron and is

$$\lambda_c = \frac{h}{m_0 c} = 0.002\ 43\text{ nm}$$

- The unshifted wavelength, λ_0 , is caused by x-rays scattered from the electrons that are tightly bound to the target atoms.

Compton Effect, Explanation

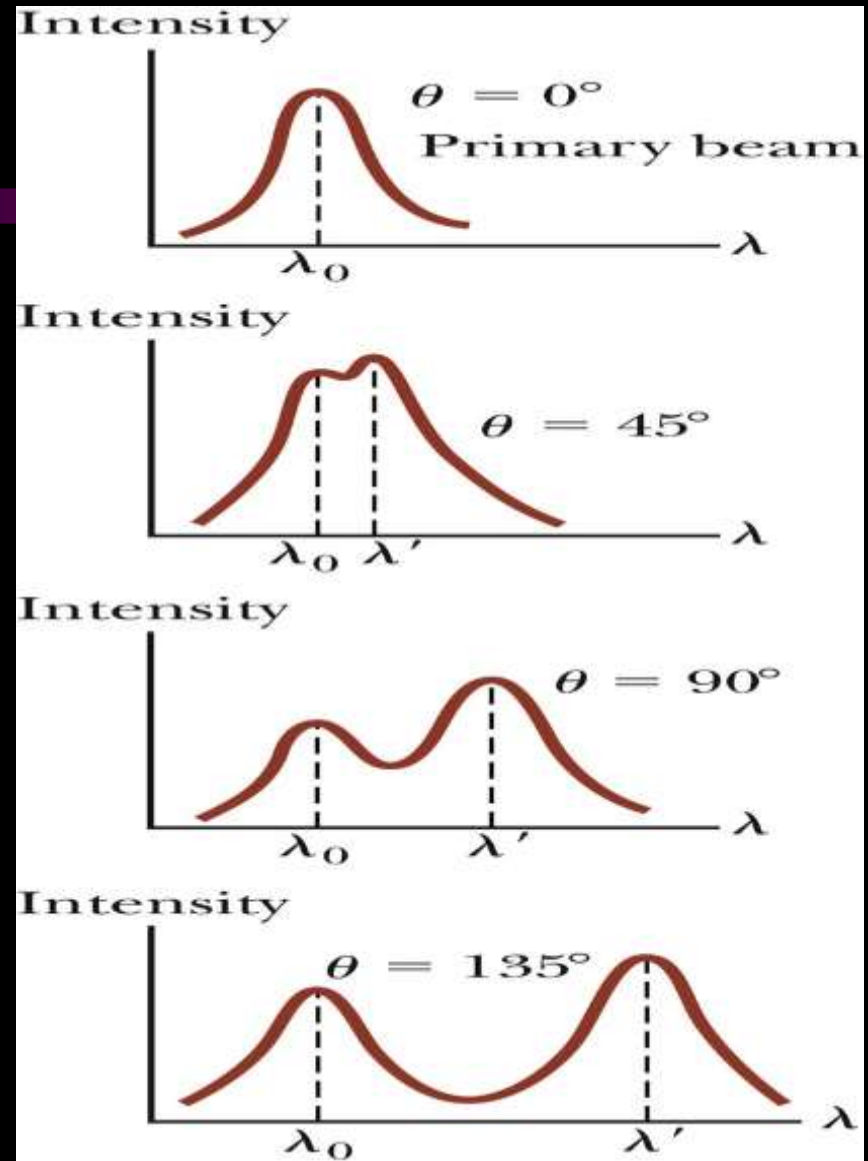
- The results could be explained by treating the photons as point-like particles having energy hf and momentum $\frac{hf}{c}$
- Assume the energy and momentum of the isolated system of the colliding photon-electron are conserved.
- This scattering phenomena is known as the **Compton effect**.

Compton Shift Equation

- The graphs show the scattered x-rays for various angles.
- The shifted peak, λ' , is caused by the scattering of free electrons.

$$\lambda' - \lambda_0 = \frac{h}{m_e c} (1 - \cos \theta)$$

- This is called the **Compton shift equation**.



Limitation Visible light can not cause Compton Effect.

As the maximum Compton shift is $0.0484 \text{ \AA}$ Hence Compton shift is insignificant for visible radiation. Therefore Compton Effect can be observed only when, the striking photon is highly energetic.

- **Scattering atom should be of low atomic no.**

For higher atomic number, the number of loosely bound electrons are less, hence the probability of modified component is less. Hence Compton shift is negligible. Therefore the intensity of unmodified component is large.

Explanation on the basis of Quantum theory

- Modified Component – Elastic Collision of X-ray photon with free or loosely bound electron.
- Unmodified Component – Collision of X-ray photon with tightly bound electron.
- In this case whole atom is involved in the collision and hence m_0 (mass of electron) in the Compton shift is replaced by M (mass of atom). As $M \gg m_0$ Compton shift is negligible. Hence unmodified components exists.

Failure of Classical Theory

- According to classical wave theory, X-rays are regarded as electromagnetic waves of frequency
- If these X-rays are incident on a material then it causes electron in the material to oscillate with the same frequency.
- These electron reradiate with same frequency in the form of electromagnetic waves .
- Hence, the scattered wave or radiation should have same wavelength as that of incident radiation.
- Secondly electron radiate uniformly in all directions.
- **Wavelength of scattered radiation should not depend on scattering angle Φ .**
- **Thus, Classical theory does not give the explanation of modified wavelength.**